

Fortinet-SD-WAN-Lab

By JoelMfungu

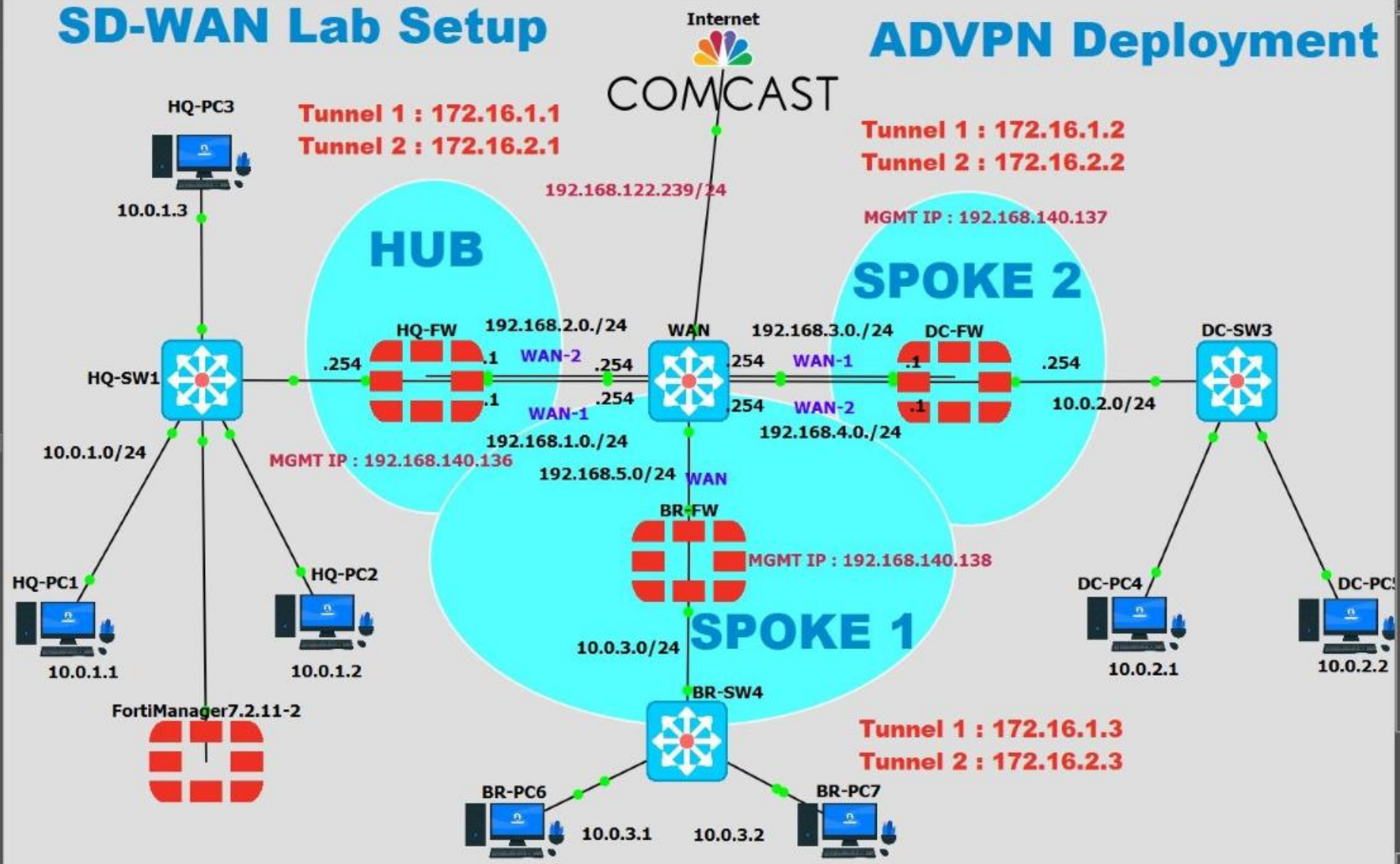
- All devices
- Filter
 - Checkpoint GAiA 80.10
 - Cisco ASAv asav-964...
 - Cisco CSR1000v 16.11...
 - Cisco IOS XRv 9000 7...
 - Cisco NX-OSv 7.3.0
 - Cloud
 - Ethernet hub
 - Ethernet switch
 - Fort
 - FortiGate 7.0.5
 - FortiManager 7.2.10
 - FortiManager 7.2.11
 - FortiWeb 7.0.3
 - Frame Relay switch
 - Internet
 - L2 Switch
 - NAT
 - PA-VM 10.0.1
 - PHILEMON
 - Routur
 - Sophos XG
 - Switch L3
 - Vbond
 - vEdge
 - vManage

+ New template

Open project

SD-WAN Lab Setup

ADVPN Deployment



Servers Summary

- GNS3 VM (GNS3) CPU ...
- MFUNGA-Pelagie CPU ...

Topology Summary

- Node
- BR-FW
 - BR-PC6
 - BR-PC7
 - BR-SW4
 - Cloud1
 - DC-FW
 - DC-PC4
 - DC-PC5
 - DC-SW3
 - FortiManager7.2.11-2
 - HQ-FW
 - HQ-PC1
 - HQ-PC2
 - HQ-PC3

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Non sécurisé 192.168.140.135/ng/interface

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Erreur

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🌐

HQ-FGT1

Dashboard

Network

Interfaces

DNS

Packet Capture

SD-WAN

Static Routes

Policy Routes

RIP

OSPF

BGP

Routing Objects

Multicast

Policy & Objects

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

FortiGate VM64-KVM

1 3 5 7 9 11 13 15 17 19 21 23

2 4 6 8 10 12 14 16 18 20 22 24

Create New

Edit

Delete

Integrate Interface

Search

Group By Type

Name	Type	Members	IP/Netmask	Administrative Access	DHCP Clients	DHCP Ranges	Re
802.3ad Aggregate 1							
fortilink	802.3ad Aggreg...		Dedicated to FortiSwitch	PING Security Fabric Connection		10.255.1.2-10.25...	2
Loopback Interface 1							
LooBACK	Loopback Interf...		10.10.10.10/255.255.255....	PING			3
Physical Interface 12							
LAN (port2)	Physical Interface		10.0.1.254/255.255.255.0	PING HTTPS SSH SNMP			3
MGMT (port1)	Physical Interface		192.168.140.135/255.255....	PING HTTPS SSH HTTP FMG-Access			C
port5	Physical Interface		0.0.0.0/0.0.0.0				C

0 Security Rating Issues

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←↻⚠ Non sécurisé 192.168.140.135/network/virtualwan?tabType=members

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admin

HQ-FGT1

Dashboard

Network

Interfaces

DNS

Packet Capture

SD-WAN

Static Routes

Policy Routes

RIP

OSPF

BGP

Routing Objects

Multicast

Policy & Objects

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

SD-WAN ZonesSD-WAN RulesPerformance SLAs

BandwidthVolumeSessions

Download

4Total

advpn1advpn2port3port4

Upload

4Total

advpn1advpn2port3port4

+ Create New

Edit

Delete

	Interfaces	Gateway	Cost	Download	Upload
	virtual-wan-link				
	SASE				
	ADVPN				
	advpn1	0.0.0.0	0	9.37 kbps	3.33 kbps
	advpn2	0.0.0.0	0	10.75 kbps	5.06 kbps
	Internet				
	WAN-1 (port3)	192.168.1.254	0	12.49 kbps	11.31 kbps
	WAN-2 (port4)	192.168.2.254	0	13.35 kbps	15.46 kbps

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admin

HQ-FGT1

Dashboard

Network

Interfaces

DNS

Packet Capture

SD-WAN

Static Routes

Policy Routes

RIP

OSPF

BGP

Routing Objects

Multicast

Policy & Objects

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

SD-WAN Zones

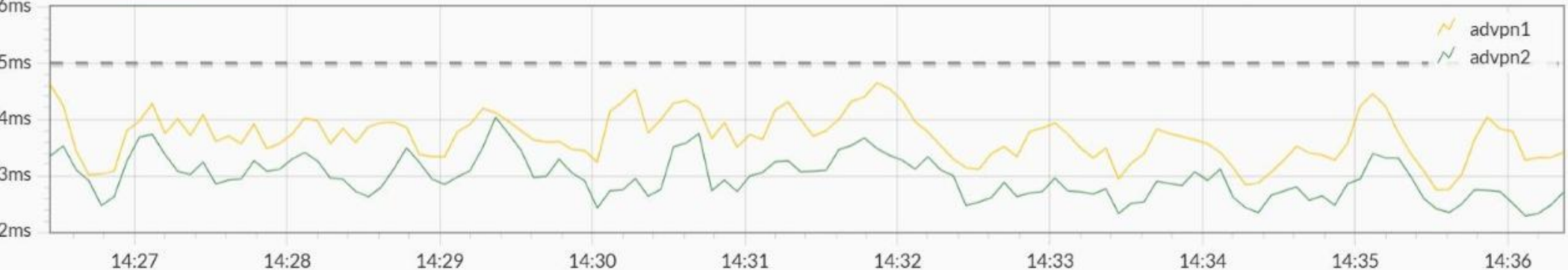
SD-WAN Rules

Performance SLAs

Packet Loss

Latency

Jitter



+ Create New

Edit

Delete

Search

Q

Name	Detect Server	Packet Loss	Latency	Jitter	Failure Threshold	Recovery Threshold
PING_BR	10.0.3.254	advpn1: 0.00% advpn2: 0.00%	advpn1: 3.41ms advpn2: 2.52ms	advpn1: 2.39... advpn2: 1.79...	5	5
PING_DC	10.0.2.254	advpn1: 0.00% advpn2: 0.00%	advpn1: 3.67ms advpn2: 3.61ms	advpn1: 2.28... advpn2: 2.32...	5	5

Fortinet

v7.0.5

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Erreur

admin

DC-FGT2

Dashboard

Network

- Interfaces
- DNS
- Packet Capture
- SD-WAN
- Static Routes
- Policy Routes
- RIP
- OSPF
- BGP
- Routing Objects
- Multicast

Policy & Objects

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

SD-WAN Zones

SD-WAN Rules

Performance SLAs

Bandwidth

Volume

Sessions

Download

Upload

Create New

Edit

Delete

	Interfaces	Gateway	Cost	Download	Upload
	virtual-wan-link				
	SASE				
	ADVPN				
	advpn1	0.0.0.0	0	4.61 kbps	1.93 kbps
	advpn2	0.0.0.0	0	6.17 kbps	1.94 kbps
	Internet				
	WAN-1 (port3)	192.168.3.254	0	13.94 kbps	14.06 kbps
	WAN-2 (port4)	192.168.4.254	0	14.66 kbps	12.93 kbps

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admin ▾

DC-FGT2 ▾

Dashboard >Network ▾

InterfacesStatic RoutesPolicy RoutesSD-WAN ☆RIPOSPFBGPRouting ObjectsMulticastPolicy & Objects >Security Profiles >VPN >User & Authentication >System 1 >Security Fabric >Log & Report >

SD-WAN ZonesSD-WAN RulesPerformance SLAs

Packet LossLatencyJitter

+ Create NewEditDelete

Search 🔍

Name ⬆	Detect Server ⬆	Packet Loss	Latency	Jitter	Failure Threshold ⬆	Recovery Threshold ⬆
PING_BR	10.0.3.254	advpn1: 🟢 100.00% advpn2: 🟢 0.00%	advpn1: 🟢 0.00ms advpn2: 🟢 7.32ms	advpn1: 🟢 0.00ms advpn2: 🟢 0.97ms	5	5
PING_HQ	10.10.10.10	advpn1: 🟢 0.00% advpn1: 🟢 0.00% advpn2: 🟢 0.00% advpn2: 🟢 0.00%	advpn1: 🟢 4.73ms advpn1: 🟢 3.55ms advpn2: 🟢 3.21ms advpn2: 🟢 3.51ms	advpn1: 🟢 3.72ms advpn1: 🟢 2.00ms advpn2: 🟢 1.99ms advpn2: 🟢 1.91ms	5	5

Fortinet v7.0.5

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Erreur

admin

HQ-FGT1

Dashboard

Network

Interfaces

DNS

Packet Capture

SD-WAN

Static Routes

Policy Routes

RIP

OSPF

BGP

Routing Objects

Multicast

Policy & Objects

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

FortiGate VM64-KVM

1 3 5 7 9 11 13 15 17 19 21 23

2 4 6 8 10 12 14 16 18 20 22 24

Create New

Edit

Delete

Integrate Interface

Search

Group By Type

	Name	Type	Members	IP/Netmask	Administrative Access	DHCP Clients	DHCP Ranges	Re
	port9	Physical Interface		0.0.0.0/0.0.0.0				C
	port10	Physical Interface		0.0.0.0/0.0.0.0				C
	WAN-1 (port3)	Physical Interface		192.168.1.1/255.255.255.0	PING HTTPS SSH			3
	advpn1	Tunnel Interface		172.16.1.1/255.255.255.255	PING			4
	WAN-2 (port4)	Physical Interface		192.168.2.1/255.255.255.0	PING HTTPS SSH			3
	advpn2	Tunnel Interface		172.16.2.1/255.255.255.255	PING			4
SD-WAN Zone 4								
	ADVPN	SD-WAN Zone	advpn1 advpn2	0.0.0.0/0.0.0.0				
	Internet	SD-WAN Zone	WAN-1 (port3) WAN-2 (port4)	0.0.0.0/0.0.0.0				
	SASE	SD-WAN Zone		0.0.0.0/0.0.0.0				

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HQ-FGT1

Dashboard

Network

Policy & Objects

Security Profiles

VPN

Overlay Controller VPN

IPsec Tunnels

IPsec Wizard

IPsec Tunnel Template

SSL-VPN Portals

SSL-VPN Settings

SSL-VPN Clients

VPN Location Map

User & Authentication

System

Security Fabric

Log & Report

☰

🔍

>

?

🔔

3

admin

+ Create New

Edit

Delete

Search

🔍

Tunnel	Interface Binding	Status	Ref.
Custom 2			
advpn1	WAN-1 (port3)	2 dialup connection(s)	4
advpn2	WAN-2 (port4)	2 dialup connection(s)	4

Fortinet

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0 Security Rating Issues

2

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⚙️

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DC-FGT2

Dashboard

Network

Policy & Objects

Security Profiles

VPN

Overlay Controller VPN

IPsec Tunnels

IPsec Wizard

IPsec Tunnel Template

SSL-VPN Portals

SSL-VPN Settings

SSL-VPN Clients

VPN Location Map

User & Authentication

System

Security Fabric

Log & Report

☰

🔍

>_

?

🔔

3

admin

+ Create New

Edit

Delete

Search

🔍

Tunnel	Interface Binding	Status	Ref.
Custom 2			
advpn1	WAN-1 (port3)	Up	4
advpn2	WAN-2 (port4)	Up	4

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 Security Rating Issues

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🏠 DC-FGT2

➤

🗺️ Dashboard

➤

🔗 Network

➤

Interfaces

DNS

Packet Capture

SD-WAN

☆

Static Routes

Policy Routes

RIP

OSPF

BGP

Routing Objects

Multicast

📁 Policy & Objects

➤

🔒 Security Profiles

➤

💻 VPN

➤

👤 User & Authentication

➤

⚙️ System

1

➤

🔗 Security Fabric

➤

📊 Log & Report

➤

☰

🔍

➤

?

🔔 3

admin

➤

SD-WAN Zones

SD-WAN Rules

Performance SLAs

+ Create New

✎ Edit

📄 Clone

🗑 Delete

Search

🔍

ID	Name	Source	Destination	Criteria	Members	Hit Count
IPv4 1						
1	PING_ADVPN	all	10.0.0.0/8	Latency	advpn1 ✓ advpn2	0
Implicit 1						
sd-wan		all	all	Source IP	any	

2

Updated: 05:38:57

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Fortinet

v7.0.5

BR-FGT3

Dashboard

Network

Policy & Objects

Security Profiles

VPN

Overlay Controller VPN

IPsec Tunnels

IPsec Wizard

IPsec Tunnel Template

SSL-VPN Portals

SSL-VPN Settings

SSL-VPN Clients

VPN Location Map

User & Authentication

System

Security Fabric

Log & Report

192.168.140.135/login?redir=%2F

192.168.140.136/login?redir=%2F

FortiGate - BR-FGT3

Non sécurisé192.168.140.137/ng/vpn/ipsec

admin

Erreur

Create New

Edit

Delete

Search

Tunnel	Interface Binding	Status	Ref.
Custom 2			
advpn1	WAN-1 (port3)	Up	4
advpn2	WAN-1 (port3)	Up	4

0 Security Rating Issues

2

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Non sécurisé 192.168.140.135/network/virtualwan?tabType=service

☆

⚙

☆

Erreur

admin

HQ-FGT1

Dashboard

Network

Interfaces

DNS

Packet Capture

SD-WAN

Static Routes

Policy Routes

RIP

OSPF

BGP

Routing Objects

Multicast

Policy & Objects

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

SD-WAN Zones

SD-WAN Rules

Performance SLAs

+ Create New

Edit

Clone

Delete

Search

ID	Name	Source	Destination	Criteria	Members	Hit Count
IPv4 2						
1	DC_Rules	all	BR- 10.0.3.0/24	Latency	advpn1 advpn2	3423
2	BR_Rules_clone	all	BR- 10.0.3.0/24	Latency	advpn1 advpn2	0
Implicit 1						
sd-wan		all	all	Source IP	any	

3

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192.168.140.135/ng/routing/bgp

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HQ-FGT1

Dashboard

Network

Interfaces

DNS

Packet Capture

SD-WAN

Static Routes

Policy Routes

RIP

OSPF

BGP

Routing Objects

Multicast

Policy & Objects

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

Local BGP Options

Local AS65000

Router ID10.0.1.254

Neighbors

Create New

Edit

Delete

IP	Remote AS
No results	

0

Neighbor Groups

Create New

Edit

Delete

Name	Remote AS
advpn1-peers	65000
advpn2-peers	65000

2

Neighbors

172.16.1.2

172.16.1.3

172.16.2.2

172.16.2.3

View Routing Monitor

Paths

5 Paths

Additional Information

API Preview

Edit in CLI

Documentation

Online Help

Video Tutorials

Apply

FORTINET

v7.0.5

BR-FGT3

Dashboard

Network

Interfaces

DNS

Packet Capture

SD-WAN

Static Routes

Policy Routes

RIP

OSPF

BGP

Routing Objects

Multicast

Policy & Objects

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

192.168.140.137/ng/interface

admin

FortiGate VM64-KVM

1 3 5 7 9 11 13 15 17 19 21 23

2 4 6 8 10 12 14 16 18 20 22 24

Create New

Edit

Delete

Integrate Interface

Search

Group By Type

	Name	Type	Members	IP/Netmask	Administrative Access	DHCP Clients	DHCP Ranges
	port9	Physical Interface		0.0.0.0/0.0.0.0			
	port10	Physical Interface		0.0.0.0/0.0.0.0			
	WAN-1 (port3)	Physical Interface		192.168.5.1/255.255.255.0	PING HTTPS SSH		
	advpn1	Tunnel Interface		172.16.1.3/255.255.255.255	PING		
	advpn2	Tunnel Interface		172.16.2.3/255.255.255.255	PING		
SD-WAN Zone 4							
	ADVPN	SD-WAN Zone	advpn1 advpn2	0.0.0.0/0.0.0.0			
	Internet	SD-WAN Zone	WAN-1 (port3)	0.0.0.0/0.0.0.0			
	SASE	SD-WAN Zone		0.0.0.0/0.0.0.0			
	virtual-wan-link	SD-WAN Zone		0.0.0.0/0.0.0.0			
Tunnel Interface 1							
	NAT interface (naf.root)	Tunnel Interface		0.0.0.0/0.0.0.0			

FortiNET

v7.0.5

0 Security Rating Issues

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Course: Fortinet NSE7 SD-WAN Tr

FortiGate - HQ-FGT1

FortiGate - DC-FGT2

192.168.140.137/login?redir=%2F

Non sécurisé192.168.140.135/ng/firewall/policy/policy/standard

HQ-FGT1

Dashboard

Network

Policy & Objects

Firewall Policy

IPv4 DoS Policy

Addresses

Internet Service Database

Services

Schedules

Virtual IPs

IP Pools

Protocol Options

Traffic Shaping

Security Profiles

VPN

User & Authentication

System

Security Fabric

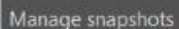
Log & Report

Create NewEditDeletePolicy LookupSearchExportInterface Pair ViewBy Sequence

Name	Source	Destination	Schedule	Service	Action	NAT	Security Profiles	Log	Bytes
ADVPN → ADVPN 1									
ADVPN to ADVPN	all	all	always	ALL	ACCEPT	Disabled	SSLno-inspection	All	873.32 kB
ADVPN → LAN (port2) 1									
ADVPN to LAN	all	all	always	ALL	ACCEPT	Disabled	SSLno-inspection	All	1.05 MB
ADVPN → LooBACK 1									
ADVPN to LooBACK	all	all	always	ALL	ACCEPT	Disabled	SSLno-inspection	UTM	6.70 MB
LAN (port2) → ADVPN 1									
LAN to ADVPN	all	all	always	ALL	ACCEPT	Disabled	SSLno-inspection	All	2.23 kB
LAN (port2) → Internet 1									
LAN to INTERNET	all	all	always	ALL	ACCEPT	Enabled	SSLno-inspection	All	0 B
LooBACK → ADVPN 1									
LoopBACK to ADVPN	all	all	always	ALL	ACCEPT	Disabled	SSLno-inspection	UTM	0 B
Implicit 1									

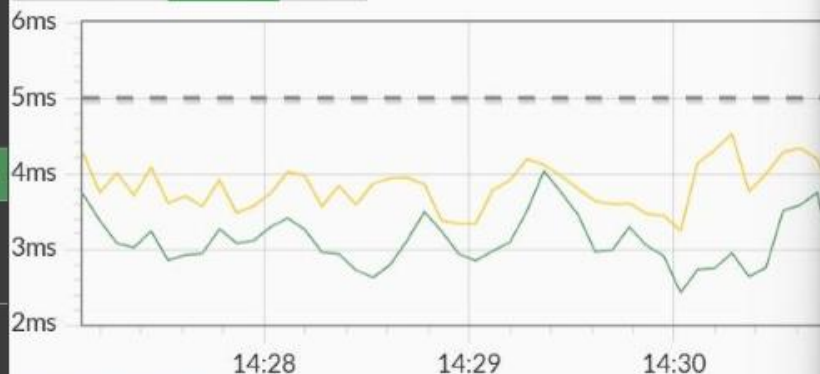
0 Security Rating Issues

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SD-WAN Zones SD-WAN Rules Performance SLAs

Packet Loss Latency Jitter



+ Create New Edit Delete Search

Name	Detect Server	Packet Loss
PING_BR	10.0.3.254	advpn1: 0.00% advpn2: 0.00%
PING_DC	10.0.2.254	advpn1: 0.00% advpn2: 0.00%

MobaXterm

Terminal Sessions View X server Tools Games Settings Macros Help

Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help X server Exit

Quick connect...

3. /home/mol 4. /home/mol 5. /home/mol

HQ-PC2 : 10.0.1.2 255.255.255.0 gateway 10.0.1.254

HQ-PC2> PING 10.0.3.1
Bad command: "PING 10.0.3.1". Use ? for help.

HQ-PC2> PING 10.0.3.1
Bad command: "PING 10.0.3.1". Use ? for help.

HQ-PC2> ping 10.0.3.1

84 bytes from 10.0.3.1 icmp_seq=1 ttl=62 time=11.161 ms
84 bytes from 10.0.3.1 icmp_seq=2 ttl=62 time=9.646 ms
84 bytes from 10.0.3.1 icmp_seq=3 ttl=62 time=6.793 ms
84 bytes from 10.0.3.1 icmp_seq=4 ttl=62 time=3.847 ms
84 bytes from 10.0.3.1 icmp_seq=5 ttl=62 time=4.945 ms

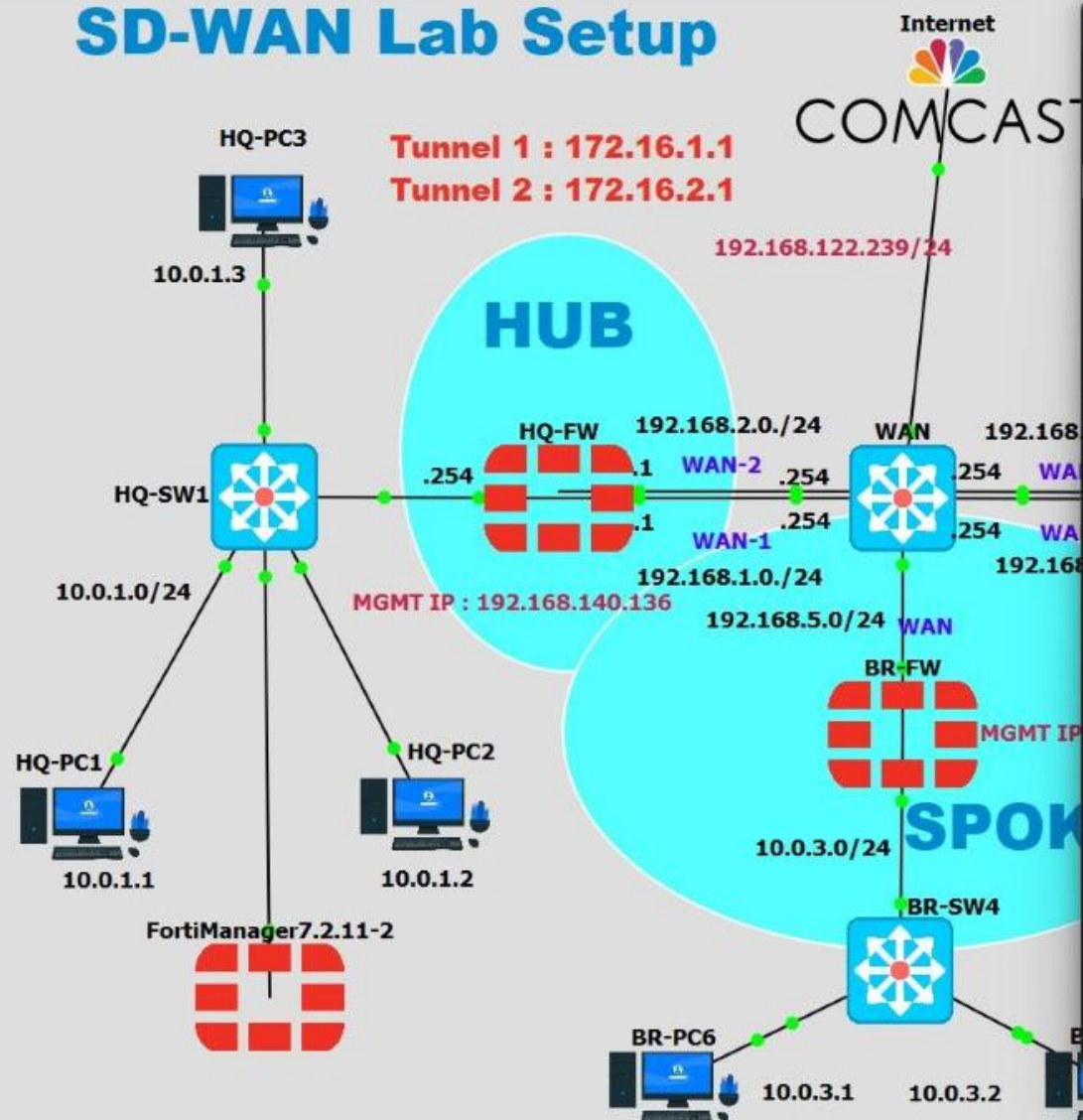
HQ-PC2> ping 10.0.3.1

84 bytes from 10.0.3.1 icmp_seq=1 ttl=62 time=5.308 ms
84 bytes from 10.0.3.1 icmp_seq=2 ttl=62 time=7.425 ms
84 bytes from 10.0.3.1 icmp_seq=3 ttl=62 time=4.251 ms
84 bytes from 10.0.3.1 icmp_seq=4 ttl=62 time=5.779 ms
84 bytes from 10.0.3.1 icmp_seq=5 ttl=62 time=12.534 ms

HQ-PC2> trace 10.0.3.1
trace to 10.0.3.1, 8 hops max, press Ctrl+C to stop
1 10.0.1.254 1.574 ms 2.478 ms 2.539 ms
2 172.16.2.3 3.126 ms 2.697 ms 1.888 ms
3 *10.0.3.1 5.642 ms (ICMP type:3, code:3, Destination port unreachable)



SD-WAN Lab Setup



MobaXterm

Terminal Sessions View Xserver Tools Games Settings Macros Help



Quick connect...

4. /home/mol x 5. /home/mol x 6. /home/mol x

Build time: Sep 9 2023 11:15:00

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Source code and license can be found at vpcs.sf.net.

For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

Checking for duplicate address...

DC-PC4 : 10.0.2.1 255.255.255.0 gateway 10.0.2.254

DC-PC4> ping 10.0.1.1

84 bytes from 10.0.1.1 icmp_seq=1 ttl=62 time=10.834 ms

84 bytes from 10.0.1.1 icmp_seq=2 ttl=62 time=14.190 ms

84 bytes from 10.0.1.1 icmp_seq=3 ttl=62 time=20.240 ms

84 bytes from 10.0.1.1 icmp_seq=4 ttl=62 time=6.025 ms

84 bytes from 10.0.1.1 icmp_seq=5 ttl=62 time=10.373 ms

DC-PC4> trace 10.0.1.1

trace to 10.0.1.1, 8 hops max, press Ctrl+C to stop

1 10.0.2.254 2.443 ms 1.332 ms 1.679 ms

2 172.16.2.1 4.626 ms 5.028 ms 3.205 ms

3 10.0.1.1 6.886 ms (ICMP type=3 code=3 Destination port unreachable)

DC-PC4>

UNREGISTERED VERSION - Please support MobaXterm by subscribing to the professional edition here: <https://mobaxterm.mobatek>



UNREGISTERED VERSION - Please support MobaXterm by subscribing to the professional edition here: <https://mobaxterm.mobatek.com/>  Manager 7.2.11-2